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THE HISTORY OF

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HYPertext

From the memex through HyperCard and the Web to peer-to-peer protocols: the long argument about how documents should link

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ACADEMIC HYPertext: MEMEX, NLS, AND THE CONFERENCES

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The founding document is Vannevar Bush's 1945 Atlantic essay "As We May Think" [1], which imagined the memex: a desk that stores a library on microfilm and lets its owner build durable associative trails between documents. Ted Nelson named the idea hypertext in the early 1960s (see the Xanadu chapter), and Douglas Engelbart built the first working system: NLS, whose 1968 "Mother of All Demos" [2] showed linking, structured documents, shared screens, and the mouse in one afternoon.

The academic tradition matured through the 1970s and 80s: Andries van Dam's HES and FRESS at Brown (students writing term papers in hypertext in 1969), ZOG at CMU, Intermedia at Brown, NoteCards at Xerox PARC, and from 1987 the ACM Hypertext conference series, which treated links, anchors, versioning, and navigation as a research field with rigor the later web largely ignored.

PRIVATE HYPERTEXT: HYPERCARD AND THE KNOWLEDGE NAVIGATOR

In 1987 hypertext went consumer. Bill Atkinson's HyperCard shipped free with every Macintosh: stacks of cards, buttons, links, and the HyperTalk scripting language readable by ordinary people. Millions of teachers, artists, and hobbyists built stacks -- the game *Myst* began as one -- making HyperCard the largest authorship explosion between BASIC and the web. The Internet Archive today preserves and emulates thousands of stacks in the browser [3]. Its limits were also instructive: stacks were single-machine and links stopped at the edge of the local disk -- private hypertext, awaiting a network.

The same year, Apple's Knowledge Navigator concept film [4] imagined the networked endpoint: a professor conversing with a software agent that follows links across the world's knowledge. Fiction, but influential fiction -- it framed the tablet, the voice assistant, and the linked global corpus as one artifact, a generation early.

CERN AND THE WEB

Tim Berners-Lee's 1989 proposal at CERN -- "Vague, but exciting", per the margin note -- yielded in 1990-91 the URL, HTTP, and HTML, and the first site at info.cern.ch [5]. The web made the engineering trade Nelson refused: links are cheap, one-way, unversioned, and allowed to break. That cheapness -- plus the 1993 decision to release the technology royalty-free -- is why it won, absorbing Gopher, WAIS, and the hypertext research agenda alike. Mosaic, Netscape, and the browser wars industrialized it; the W3C's own short history [6] tells the story from inside.

What was lost is the standing complaint of this whole series: bidirectional links, transclusion, versioning, provenance, and

authorship as a first-class right rather than a platform privilege.

P2P HYPERTEXT: SCUTTLEBUTT, DAT, BEAKER, IPFS, BLUESKY

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The 2010s produced a counter-tradition: hypertexts that are internet-adjacent but not web-standard, distributing documents peer-to-peer with cryptographic identity instead of servers. Secure Scuttlebutt [7] builds a social web from append-only signed logs that replicate through gossip -- famously usable off-grid, sneakernet included. The Dat protocol [8] (now the Hypercore stack) gives versioned, syncable archives addressed by key; Beaker Browser [9] (2016-2022) made Dat browsable and writable -- a personal web server in every browser tab, the closest anyone has come to HyperCard-on-the-network. IPFS [10] provides content-addressed storage where a document's address is the hash of its content -- Xanadu's permanent documents, rebuilt on Merkle DAGs.

Bluesky's AT Protocol [11] carries the line into social media: user data lives in signed, portable repositories; identity is a DID you can move between hosts; the network is an open commons of feeds and moderation services rather than one company's database. Each system trades away some of the web's universality to recover properties the academic tradition prized -- provenance, permanence, user ownership. The argument Bush started in 1945 is still open, and still productive.

SOURCES & CITATIONS

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- [1] Vannevar Bush, As We May Think, The Atlantic, July 1945
<https://www.theatlantic.com/magazine/archive/1945/07/as-we-may-think/3>
- [2] The Mother of All Demos, presented by Douglas Engelbart

- (1968), on YouTube; context at the Doug Engelbart Institute, dougengelbart.org
<https://www.youtube.com/watch?v=yJDv-zdhzMY>
- [3] HyperCard on the Archive: emulated HyperCard stacks collection, Internet Archive
<https://archive.org/details/hypercardstacks>
- [4] Apple, Knowledge Navigator concept video (1987), DigiBarn copy on YouTube
<https://www.youtube.com/watch?v=-jiBLQyUi38>
- [5] The birthplace of the Web: info.cern.ch, home of the first website (restored)
<https://info.cern.ch/>
- [6] W3C, A Little History of the World Wide Web
<https://www.w3.org/History.html>
- [7] Secure Scuttlebutt: protocol guide and community
<https://scuttlebutt.nz/>
- [8] Dat protocol / Hypercore: versioned peer-to-peer data
<https://dat-ecosystem.org/>
- [9] Beaker Browser: an experimental peer-to-peer web browser (archived project)
<https://github.com/beakerbrowser/beaker>
- [10] Juan Benet, IPFS: Content Addressed, Versioned, P2P File System, 2014
<https://arxiv.org/abs/1407.3561>
- [11] The AT Protocol (Bluesky), specifications and documentation
<https://atproto.com/>

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